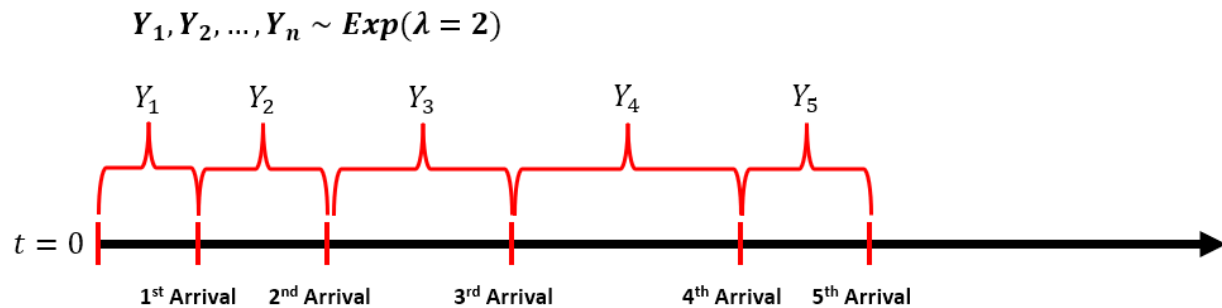


# Question 1

M2L12



$X = \text{Time between 3<sup>rd</sup> and 5<sup>th</sup> arrivals}$

$$X = Y_4 + Y_5 \sim \text{Erlang}_2(\lambda = 2)$$

See [2018 Practice Exam 1 Additional Notes](#) Question 8 and [2019 Practice Exam 1 Additional Notes](#) Question 12 to review the calculations for the CDF of the Erlang distribution.

Review Sections 4.2.2.1 in Goldsman, D. & Goldsman, P. (2020). A First Course in Probability and Statistics. Lulu. [Download link](#)

**Theorem:** Let  $X$  be the amount of time until the *first arrival* of a Poisson process with rate  $\lambda$ . Then  $X \sim \text{Exp}(\lambda)$ .

**Proof:** The cdf of  $F(x)$  is

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= 1 - P(\text{no arrivals in } [0, x]) \\ &= 1 - \frac{e^{-\lambda x} (\lambda x)^0}{0!} \quad (\text{the number of arrivals in } [0, x] \text{ is } \text{Pois}(\lambda x)) \\ &= 1 - e^{-\lambda x}. \quad \square \end{aligned}$$

**Theorem:** Amazingly, it can be shown (after a lot of work) that the interarrival times of a Poisson process are *all* iid  $\text{Exp}(\lambda)$ ! See for yourself when you take an advanced stochastic processes course.

## Question 2

M2L11

The calculation of  $E[XY]$  is using the bivariate version of LOTUS with  $h(X, Y) = XY$

**“Definition”** (two-dimensional LOTUS): Suppose that  $h(X, Y)$  is some function of the RV's  $X$  and  $Y$ . Then

$$E[h(X, Y)] = \begin{cases} \sum_x \sum_y h(x, y) f(x, y) & \text{if } (X, Y) \text{ is discrete} \\ \int_{\mathbb{R}} \int_{\mathbb{R}} h(x, y) f(x, y) dx dy & \text{if } (X, Y) \text{ is continuous} \end{cases}$$

## Question 3

M4L3

# Even More Remarks

Be careful about events that take place at the same time, e.g., a guy shows up at the same time that another guy finishes getting served. Simply establish ground rules for how to deal with ties.

Every discrete-event simulation language maintains a FEL somewhere deep in its cold, cold heart.

## Question 4

M5L9

# Variables

- Unlike attributes, whose values are specific to each customer, **variables are global**.
- If you change a variable anyplace in the Arena program, it gets changed everywhere.
- Example: WIP might be incremented when an entity is Create'd, and decremented if it's Dispose'd – these events could occur anywhere.

## Question 5

M5L17

# Seize, Delay, Release

- Why do we have separate Seize, Delay, and Release modules? Aren't they right there in the Process module?
- Yup, but sometimes you need to do things that are too complicated for the Process module version of Seize–Delay–Release.
- Now you can handle things like...
  - Seize–Assign–Delay–Release
  - Non-symmetric multiple Seize's and Release's
  - Complicated Seize's and Release's that might depend on sets of servers (as in the upcoming Call Center example)

## Question 6

M5L6 M5L17 M5L19

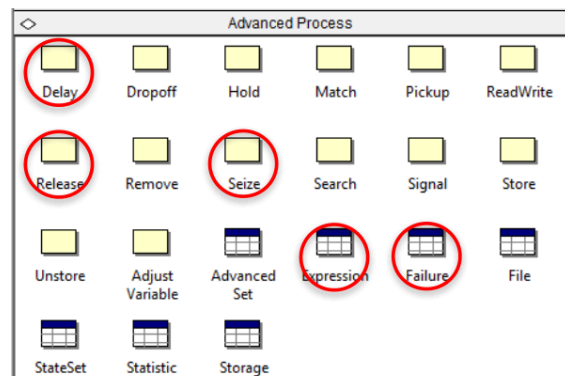
# Seize-Delay-Release

You'll take one of the following "actions" in the Process module:

- **Delay**: Spend time in the Process (self-service).
- **Seize-Delay-Release**: Grab at least one resource (server), spend time getting served, and then free the server for the next customer. If you Seize and the server isn't available, you may have to wait in a **queue**. [www.youtube.com/watch?v=eAyVgPtC5Zc](http://www.youtube.com/watch?v=eAyVgPtC5Zc)
- **Seize-Delay**: Grab at least one resource and spend time getting served. Remember to Release the server sometime later, else he gets deadlocked and a giant line will form!
- **Delay-Release**: Use a previously Seize'd server for a while, and then free him for the next guy to use.

## Advanced Process Template

- File > Template Panel > Attach (and hopefully you find it in the Rockwell Software \Arena \ Template directory).
- Lots of new modules and spreadsheets.
- I don't expect you to memorize them all.
- Concentrate for now on a few of these (comment on others later):
  - Seize, Delay, Release modules
  - Expression, Failure spreadsheets.



# Blocks Template

- File > Template Panel > Attach (and hopefully you find it in the Rockwell Software \ Arena \ Template directory).

- Huge number of blocks!
- Do not be scared!
- They're sort of a self-contained language (related to SIMAN).
- Very specialized and low-level.
- Very useful, but we'll only need a few for now:
  - Seize, Delay, Release, Queue, Alter

| Blocks                              |                                    |                                   |                                    |
|-------------------------------------|------------------------------------|-----------------------------------|------------------------------------|
| <input type="checkbox"/> Access     | <input type="checkbox"/> Activate  | <input type="checkbox"/> Allocate | <input type="checkbox"/> Alter     |
| <input type="checkbox"/> Assign     | <input type="checkbox"/> Begin     | <input type="checkbox"/> Block    | <input type="checkbox"/> Branch    |
| <input type="checkbox"/> Capture    | <input type="checkbox"/> Close     | <input type="checkbox"/> Combine  | <input type="checkbox"/> Convey    |
| <input type="checkbox"/> Copy       | <input type="checkbox"/> Count     | <input type="checkbox"/> Create   | <input type="checkbox"/> Delay     |
| <input type="checkbox"/> Detect     | <input type="checkbox"/> Dispose   | <input type="checkbox"/> Dropoff  | <input type="checkbox"/> Duplicate |
| <input type="checkbox"/> Else       | <input type="checkbox"/> Elself    | <input type="checkbox"/> EndIf    | <input type="checkbox"/> EndWhile  |
| <input type="checkbox"/> Event      | <input type="checkbox"/> Exit      | <input type="checkbox"/> FindJ    | <input type="checkbox"/> Free      |
| <input type="checkbox"/> Group      | <input type="checkbox"/> Halt      | <input type="checkbox"/> If       | <input type="checkbox"/> Include   |
| <input type="checkbox"/> Insert     | <input type="checkbox"/> Match     | <input type="checkbox"/> Move     | <input type="checkbox"/> OnChange  |
| <input type="checkbox"/> PickQ      | <input type="checkbox"/> Pickup    | <input type="checkbox"/> Preempt  | <input type="checkbox"/> Proceed   |
| <input type="checkbox"/> QPick      | <input type="checkbox"/> Queue     | <input type="checkbox"/> Read     | <input type="checkbox"/> Release   |
| <input type="checkbox"/> Relinquish | <input type="checkbox"/> Remove    | <input type="checkbox"/> Request  | <input type="checkbox"/> Route     |
| <input type="checkbox"/> Scan       | <input type="checkbox"/> Search    | <input type="checkbox"/> Seize    | <input type="checkbox"/> Select    |
| <input type="checkbox"/> Signal     | <input type="checkbox"/> Split     | <input type="checkbox"/> Stack    | <input type="checkbox"/> Start     |
| <input type="checkbox"/> Station    | <input type="checkbox"/> Stop      | <input type="checkbox"/> Store    | <input type="checkbox"/> Tally     |
| <input type="checkbox"/> Trace      | <input type="checkbox"/> Transport | <input type="checkbox"/> Unblock  | <input type="checkbox"/> Unstore   |
| <input type="checkbox"/> VBA        | <input type="checkbox"/> Wait      | <input type="checkbox"/> While    | <input type="checkbox"/> Write     |
| <input type="checkbox"/> Zap        |                                    |                                   |                                    |

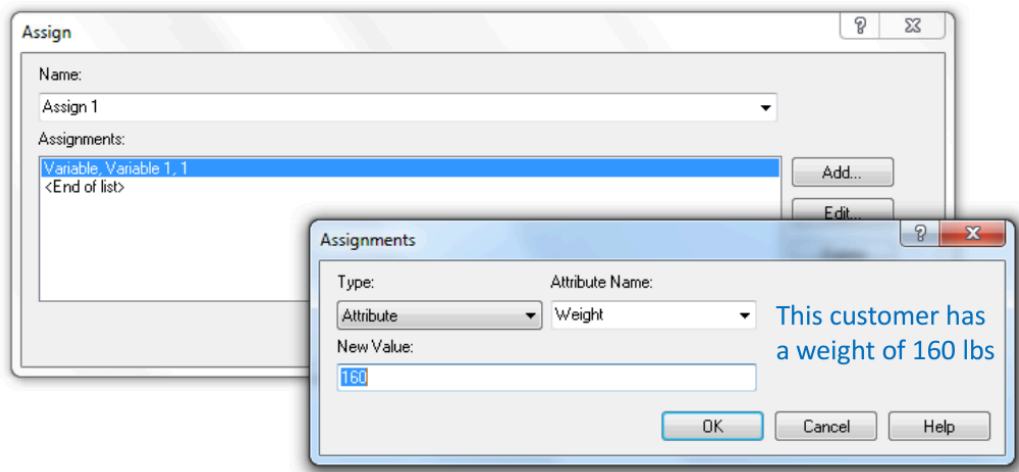
## Question 7

M5L9

# Assign Module

Assign

- So how can you change attributes, variables, and other stuff?
- Use the very powerful and flexible [Assign](#) module.



## Question 8

M5L20

# Resource Sets

- Use the Set spreadsheet in the Basic Process template to define sets.
- We'll just study resource sets for now.
- A “vanilla” **resource** has identical, interchangeable servers. But a **resource set** can have **distinct** servers, with different schedules, service speeds, service specialties, etc.
- Call Center Example: Three products with the following resources
  - Product 1: Charity, Noah, Molly, Anna, Sammy
  - Product 2: Tierney, Sean, Emma, Anna, Sammy
  - Product 3: Shelley, Jenny, Christie, Molly, Anna, Sammy
  - Molly, Anna, and Sammy all have degree of cross-functionality.
  - All 11 servers turn out to have different schedules (more on that later)



## Question 9

M5L8

# Decide Module



Decide

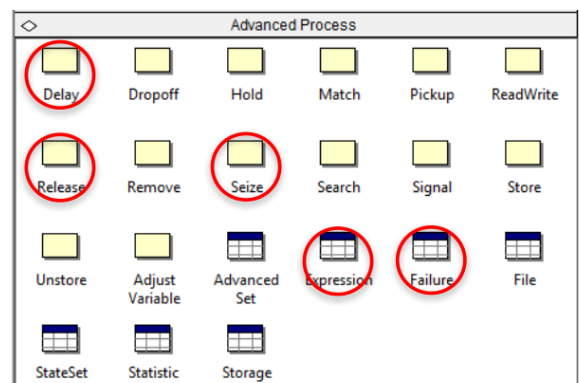
- When an entity gets to a Decide module, he can...
  - Randomly go to either of two locations (“**2-way by Chance**”) (probabilities are given in percentages)
  - Randomly go to any of various locations (“**N-way by Chance**”)
  - Go to either of two locations if a condition is satisfied (“**2-way by Condition**”)
  - Go to any of various locations if a condition is satisfied (“**N-way by Condition**”)

## Question 10

M5L17

# Advanced Process Template

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  - Seize, Delay, Release modules
  - Expression, Failure spreadsheets.



## Question 11

M5L6

# Seize-Delay-Release

You'll take one of the following "actions" in the Process module:

- **Delay:** Spend time in the Process (self-service).
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- **Seize-Delay:** Grab at least one resource and spend time getting served. Remember to Release the server sometime later, else he gets deadlocked and a giant line will form!
- **Delay-Release:** Use a previously Seize'd server for a while, and then free him for the next guy to use.

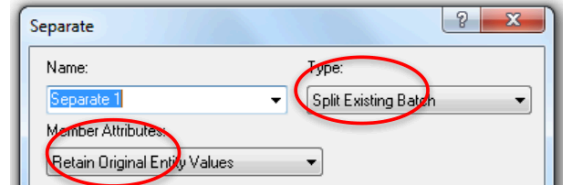
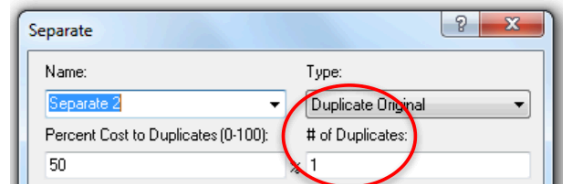
## Question 12

M5L13

# Separate Module

Separate

- Duplicate a single entity, or split multiple entities that had been combined in a Batch module.
- If dealing with a **Permanent** batch, usually use Duplicate Original to get several customers all with the same attributes.
- If dealing with a **Temporary** batch, use Split Existing Batch to reproduce customers with original attributes.



## Question 13

M5L27 M5L28 M5L29

# Description

- Manufacturing cell with 3 types of parts.
- Each part type follows a different path (“**sequence**”) thru the system.
  - Different service times at each station (depending on part type and place in visitation sequence).
  - E.g., Part type 2’s visit stations 1 > 2 > 4 > 2 > 3, each with different service times.
- Movement requires the Advanced Transfer template
  - Route, Station, Enter, Leave modules
  - Sequences spreadsheet
  - “Advanced Sets” (which is where the sequences come in).
- Parts can move in a variety of ways:
  - By themselves, transporters, conveyors.
  - Requires construction of transporter and conveyor paths

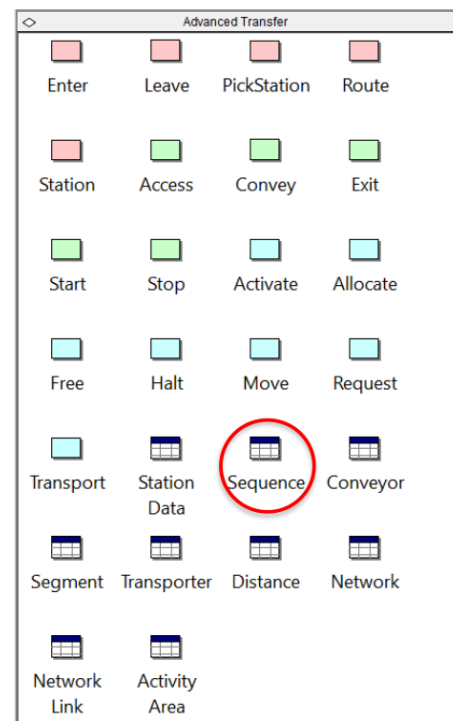
## Sequences

Each type of customer follows a different path (**sequence**) thru the system.

- A sequence goes from station to station
- Maybe have different service time distribution at each station (depending on customer type and place in visitation sequence).

Example:

- Part Type 1’s visit stations 1 > 2 > 3 > 4
- Part Type 2’s visit 1 > 2 > 4 > 2 > 3
- Part Type 3’s visit 2 > 1 > 3
- Each with different service times distributions

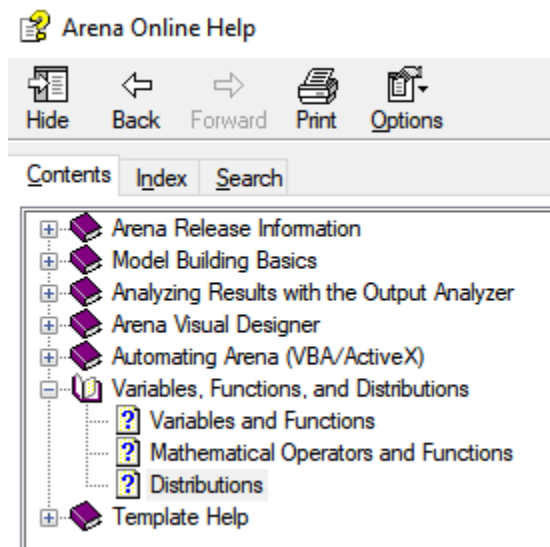
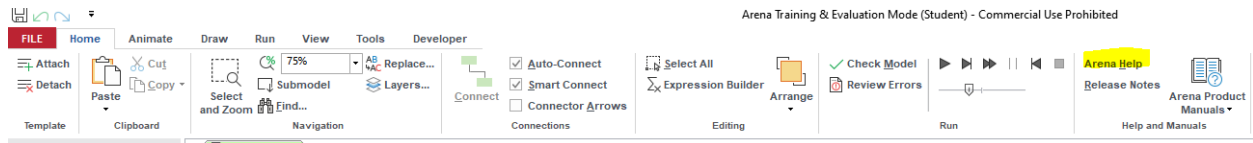


# Question 14

M2L7 M5L23

The documentation for DISC is instructive on how to use this function.

You can either search for it, or go to Variables, Functions, and Distributions -> Distributions.



## Empirical DISCRETE Distribution

---

DISCcrete(Prob1,Value1,Prob2,Value2, . . . [,Stream])

Prob<sub>i</sub> — Cumulative probability

Value<sub>i</sub> — Value associated with probability Prob<sub>i</sub>

Parameter Set — Parameter set ID containing values for Prob<sub>i</sub> and Value<sub>i</sub> pairs

When evaluating the DISCcrete function, [SIMAN](#) generates a random number between 0 and 1. SIMAN then searches the probabilities, Prob<sub>i</sub>, until it finds one larger than the random number. The Value<sub>i</sub> associated with that probability is returned.

Example: DISC(0.3,1, 0.8,2, 1.0,3) This function returns the value 1 approximately 30 percent of the time, the value 2 approximately 50 percent of the time (0.8 - 0.3 = 0.5 or 50 percent), and the value 3 approximately 20 percent of the time (1.0 - 0.8 = 0.2 or 20 percent).

Drawing a random value from the empirical discrete distribution uses a sequential search through the cumulative probabilities table. Let (P<sub>k</sub>, V<sub>k</sub>) be a (cumulative probability, value) pair. The search algorithm proceeds as follows:

```
Initialize k=1
Generate U ~ U( 0,1 )
if U < Pk, then
    DISCRETE = Vk
and return
Otherwise, increment k and repeat check.
```

Note that the value returned is the lowest V<sub>k</sub> value associated with the first cumulative probability, P<sub>k</sub>, that is greater than the sample drawn from the uniform distribution. This contrasts with the empirical continuous distribution which interpolates between probability values to obtain a random value.

For the function in this question, we have DISC(0.5, -1, 1, 1). This corresponds to the following CDF:

$$F(x) = \begin{cases} 0 & x < -1 \\ 0.5 & -1 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Using the example in the documentation, we can deduce that this will return the value -1 50% of the time and the value 1 (1 - 0.5 = 0.5) 50% of the time.

And the corresponding PDF:

$$f(x) = \begin{cases} 0.5 & x = -1 \\ 0.5 & x = 1 \\ 0 & \text{otherwise} \end{cases}$$

The expected value of this random variable is  $E[X] = \sum_x x f(x) = (-1)(0.5) + (1)(0.5) = 0$

$$E[X^2] = \sum_x x^2 f(x) = (-1)^2(0.5) + (1)^2(0.05) = 0.5 + 0.5 = 1$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 1 - 0 = 1$$

The expression  $(1 == 1)$  evaluates to 1 (true).

We want to find the variance of  $2X + 1$ .

$$\text{Var}(2X + 1) = 4\text{Var}(X) = 4(1) = 4$$

## Question 15

### M6L2

#### c. Mid-Square Method (J. von Neumann)

Idea: Take the middle part of the square of the previous random number. von Neumann was a fun-loving guy, but method is terrible!

Example: Take  $R_i = X_i/10000, \forall i$ , where  $X_i$ 's are integers  $< 10000$ .

Set *seed*  $X_0 = 6632$ ; then  $6632^2 \rightarrow 43983424$ ;

So  $X_1 = 9834$ ; then  $9834^2 \rightarrow 96707556$ ;

So  $X_2 = 7075$ , etc,...

Unfortunately, positive serial correlation in  $R_i$ 's.

Also, occasionally degenerates; e.g., consider  $X_i = 0003$ .



## Question 16

M1L8 M2L6 M3L8 M6L3

# Lesson Overview

Last Time: Discussed some awful  $\text{Unif}(0,1)$  generators.

This Time: We'll look at a really good one – the *linear congruential generator*.

Variations of the LCG are the most commonly used generators in practice.



## Question 17

M6L3 M6L6

$$X_0 = 3$$

$$X_1 = 11 \bmod 5 = 1$$

$$X_2 = 5 \bmod 5 = 0$$

$$X_3 = 2 \bmod 5 = 2$$

$$X_4 = 8 \bmod 5 = 3$$

The period is 4.

## Question 18

### M6L3 M6L6

$c = 124$  and  $m = 2048$ ; these are not [relatively prime](#) because they share a common factor of 4.

Other lines of reasoning can be used as well.

We need to see whether the LCG actually manages to produce all the  $m$  values. Let's consider how the LCG works. When we start with an initial seed value (say  $X_0$ ), the next value is computed as  $(32X_0 + 124) \bmod 2048$ . Every subsequent value depends only on the previous value.

Given that  $a$  and  $c$  are both even numbers, we can infer that the output of the generator,  $X_i$ , will also be even.

The multiplication of two even numbers always results in another even number. This is because even numbers can be expressed in the form  $2k$ , where  $k$  is any integer. When we multiply two even numbers, the resulting product can also be expressed in the form  $2k$ .

The sum of two even numbers is always even. This is because when we add two even numbers together, their combined form is still expressible as  $2k$ , where  $k$  is an integer.

## Question 19

### M1L8 M2L6 M3L8 M6L3

**Better Example** (desert island generator): Here's our old 16807 implementation (BFS 1987), which I've translated from FORTRAN. It works fine, is fast, and is full-period with cycle length  $> 2$  billion,

$$X_i = 16807 X_{i-1} \bmod (2^{31} - 1).$$

## Question 20

M6L4

How do we go from  $B_i$ 's to  $\text{Unif}(0,1)$ 's?

Easy way: Use ( $\ell$ -bits in base 2)/ $2^\ell$  and convert to base 10.

**Example:** Set  $\ell = 4$  in previous example and get:

$$1111_2, 1000_2, 1101_2, 1101_2, \dots \rightarrow \frac{15}{16}, \frac{8}{16}, \frac{13}{16}, \frac{13}{16}, \dots \quad \square$$

## Question 21

M6L8

In fact, we *reject* the null hypothesis  $H_0$  (that the observations are uniform) if  $\chi_0^2 > \chi_{\alpha, k-1}^2$ , where  $\chi_{\alpha, k-1}^2$  is the appropriate  $(1 - \alpha)$  quantile from a  $\chi^2$  table, i.e.,  $P(\chi_{k-1}^2 < \chi_{\alpha, k-1}^2) = 1 - \alpha$ .

If  $\chi_0^2 \leq \chi_{\alpha, k-1}^2$ , we *fail to reject*  $H_0$ .

Also see [How to use the Normal/t/Chi2 Tables](#)

## Question 22

M6L9

Let  $A$  denote the total number of runs “up and down” out of  $n$  observations. ( $A = 6$  in the above example.)

Amazing Fact: If  $n$  is large (say,  $\geq 20$ ) and the  $R_j$ 's are actually independent, then

$$A \approx \text{Nor}\left(\frac{2n-1}{3}, \frac{16n-29}{90}\right).$$



We'll reject  $H_0$  if  $A$  is too big or small. The test statistic is

$$Z_0 = \frac{A - E[A]}{\sqrt{\text{Var}(A)}},$$

and we reject  $H_0$  if  $|Z_0| > z_{\alpha/2}$ .

## Question 23

M7L2 M7L6

**Example:**  $\text{Erlang}_n(\lambda)$ . If  $X_1, \dots, X_n \sim \text{i.i.d. Exp}(\lambda)$ , then  $Y = \sum_{i=1}^n X_i \sim \text{Erlang}_n(\lambda)$ . By inverse transform,

$$Y = \sum_{i=1}^n X_i = \sum_{i=1}^n \left[ \frac{-1}{\lambda} \ln(U_i) \right] = \frac{-1}{\lambda} \ln\left(\prod_{i=1}^n U_i\right).$$

This only takes one natural log evaluation, so it's pretty efficient.

## Question 24

### M2L12 M7L2

If  $X$  has distribution  $\Phi(x)$  this means that  $X$  is a standard Normal random variable. The CDF for the [standard Normal gets this special character](#) of  $\Phi$  since it is so special and widely used.

The random variable  $F(X)$  (or in the specific case of the standard Normal distribution,  $\Phi(x)$ ) is  $\text{Unif}(0, 1)$ . This is a direct statement of the inverse transform theorem.

## Question 25

### M2L5 M2L12 M7L2

We set  $\Phi(X) = U$  and solve for  $X$  by taking the inverse of each side.

This results in  $X = \Phi^{-1}(U)$  which means that  $X$  will have a standard Normal distribution.

This is basically saying that if we substitute uniform random numbers between 0 and 1 into the inverse CDF of the standard Normal distribution, the result that is generated will be random variates from the standard Normal distribution.

So we know that  $X = \Phi^{-1}(U) \sim \text{Nor}(0, 1)$

$$E[2X + 3] = 2E[X] + 3 = 2(0) + 3 = 3$$

$$\text{Var}(2X + 3) = 4\text{Var}(X) = 4(1) = 4$$

$$2\Phi^{-1}(U) + 3 \sim \text{Nor}(3, 4)$$

## Question 26

### M7L2

Recall that  $-\frac{1}{\lambda} \ln(1 - U) \sim -\frac{1}{\lambda} \ln(U) \sim \text{Exp}(\lambda)$

Using properties of logarithms:

$$-3[\ln(U^2) + \ln(V^2)] = -3[2\ln(U) + 2\ln(V)] = -6\ln(U) + -6\ln(V)$$

This is summing two iid exponential random variables with  $\lambda = 1/6$  which is an Erlang distribution with  $n = 2$  and  $\lambda = 1/6$

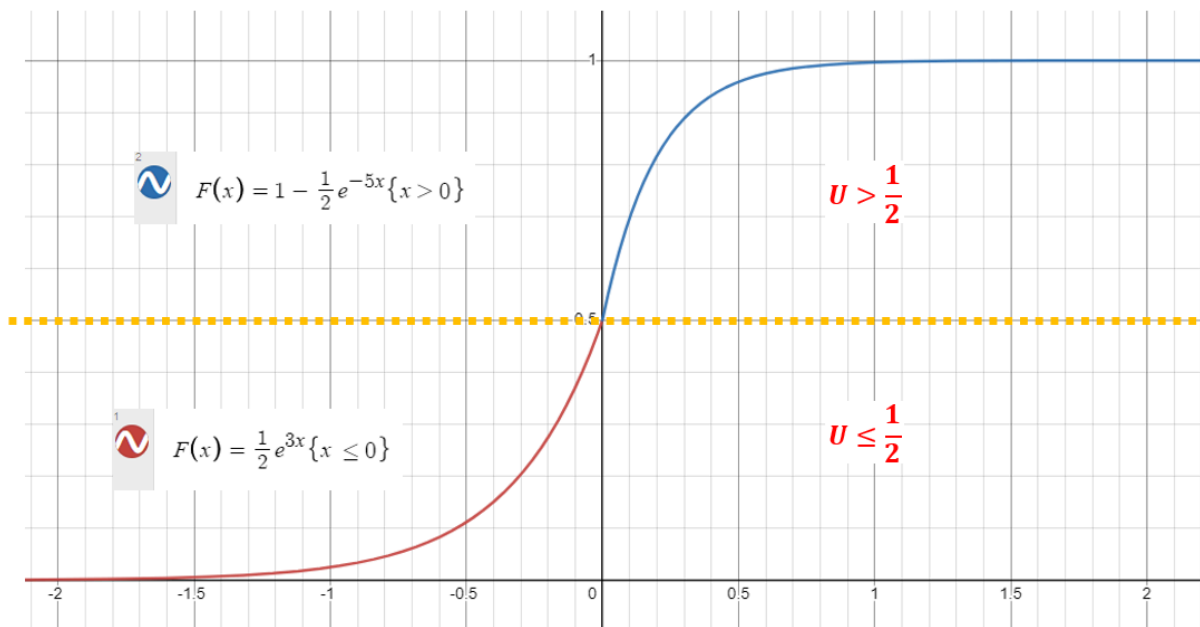
## Question 27

### M7L2 M7L3

To find the cut off point for the values of  $U$ , we can do it analytically or look at a graph.

Analytically, we would need to look at the point  $x = 0$  (since this is where the CDF changes over).

When  $x = 0$ ,  $\frac{1}{2}e^0 = \frac{1}{2}$ , so for values  $\leq 0$ , we would map  $U$  values  $\leq \frac{1}{2}$



For  $U \leq 1/2$ ,

$$U = \frac{1}{2}e^3 X$$

$$2U = e^3 X$$

$$\ln(2U) = 3X$$

$$X = \frac{1}{3} \ln(2U)$$

For  $U > 1/2$

$$U = 1 - \frac{1}{2} e^{-5X}$$

$$\frac{1}{2} e^{-5X} = 1 - U$$

$$e^{-5X} = 2(1 - U)$$

$$-5X = \ln(2(1 - U))$$

$$X = -\frac{1}{5} \ln(2(1 - U))$$

## Question 28

M7L2 M7L3

Since  $U > 1/2$ , we use the second piece of the inverse CDF.

## Question 29

M7L12

**Box–Muller Method:** Here's a nice, easy way to generate standard normals.

**Theorem:** If  $U_1, U_2$  are i.i.d.  $\mathcal{U}(0,1)$ , then

$$\begin{aligned} Z_1 &= \sqrt{-2\ln(U_1)} \cos(2\pi U_2) \\ Z_2 &= \sqrt{-2\ln(U_1)} \sin(2\pi U_2) \end{aligned}$$

are i.i.d.  $\text{Nor}(0,1)$ .

## Question 30

M7L12 M7L13 M7L6

Recall

$\chi^2(n)$  distribution: If  $Z_1, Z_2, \dots, Z_n$  are i.i.d.  $\text{Nor}(0,1)$ , then  $\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$ .

Since the table gives us probabilities to the right, you can think of what is being asked as

$$1 - 0.99 = 1 - P(\chi^2(3) < c)$$

$$0.01 = P(\chi^2(3) \geq c)$$

This is found by using the table:



| Percentage Points of the Chi-Square Distribution |                                           |        |        |        |        |       |       |       |       |
|--------------------------------------------------|-------------------------------------------|--------|--------|--------|--------|-------|-------|-------|-------|
| Degrees of Freedom                               | Probability of a larger value of $\chi^2$ |        |        |        |        |       |       |       |       |
|                                                  | 0.99                                      | 0.95   | 0.90   | 0.75   | 0.50   | 0.25  | 0.10  | 0.05  | 0.01  |
| 1                                                | 0.000                                     | 0.004  | 0.016  | 0.102  | 0.455  | 1.32  | 2.71  | 3.84  | 6.63  |
| 2                                                | 0.020                                     | 0.103  | 0.211  | 0.575  | 1.386  | 2.77  | 4.61  | 5.99  | 9.21  |
| 3                                                | 0.115                                     | 0.352  | 0.584  | 1.212  | 2.366  | 4.11  | 6.25  | 7.81  | 11.34 |
| 4                                                | 0.297                                     | 0.711  | 1.064  | 1.923  | 3.357  | 5.39  | 7.78  | 9.49  | 13.28 |
| 5                                                | 0.554                                     | 1.145  | 1.610  | 2.675  | 4.351  | 6.63  | 9.24  | 11.07 | 15.09 |
| 6                                                | 0.872                                     | 1.635  | 2.204  | 3.455  | 5.348  | 7.84  | 10.64 | 12.59 | 16.81 |
| 7                                                | 1.239                                     | 2.167  | 2.833  | 4.255  | 6.346  | 9.04  | 12.02 | 14.07 | 18.48 |
| 8                                                | 1.647                                     | 2.733  | 3.490  | 5.071  | 7.344  | 10.22 | 13.36 | 15.51 | 20.09 |
| 9                                                | 2.088                                     | 3.325  | 4.168  | 5.899  | 8.343  | 11.39 | 14.68 | 16.92 | 21.67 |
| 10                                               | 2.558                                     | 3.940  | 4.865  | 6.737  | 9.342  | 12.55 | 15.99 | 18.31 | 23.21 |
| 11                                               | 3.053                                     | 4.575  | 5.578  | 7.584  | 10.341 | 13.70 | 17.28 | 19.68 | 24.72 |
| 12                                               | 3.571                                     | 5.226  | 6.304  | 8.438  | 11.340 | 14.85 | 18.55 | 21.03 | 26.22 |
| 13                                               | 4.107                                     | 5.892  | 7.042  | 9.299  | 12.340 | 15.98 | 19.81 | 22.36 | 27.69 |
| 14                                               | 4.660                                     | 6.571  | 7.790  | 10.165 | 13.339 | 17.12 | 21.06 | 23.68 | 29.14 |
| 15                                               | 5.229                                     | 7.261  | 8.547  | 11.037 | 14.339 | 18.25 | 22.31 | 25.00 | 30.58 |
| 16                                               | 5.812                                     | 7.962  | 9.312  | 11.912 | 15.338 | 19.37 | 23.54 | 26.30 | 32.00 |
| 17                                               | 6.408                                     | 8.672  | 10.085 | 12.792 | 16.338 | 20.49 | 24.77 | 27.59 | 33.41 |
| 18                                               | 7.015                                     | 9.390  | 10.865 | 13.675 | 17.338 | 21.60 | 25.99 | 28.87 | 34.80 |
| 19                                               | 7.633                                     | 10.117 | 11.651 | 14.562 | 18.338 | 22.72 | 27.20 | 30.14 | 36.19 |
| 20                                               | 8.260                                     | 10.851 | 12.443 | 15.452 | 19.337 | 23.83 | 28.41 | 31.41 | 37.57 |
| 22                                               | 9.542                                     | 12.338 | 14.041 | 17.240 | 21.337 | 26.04 | 30.81 | 33.92 | 40.29 |
| 24                                               | 10.856                                    | 13.848 | 15.659 | 19.037 | 23.337 | 28.24 | 33.20 | 36.42 | 42.98 |
| 26                                               | 12.198                                    | 15.379 | 17.292 | 20.843 | 25.336 | 30.43 | 35.56 | 38.89 | 45.64 |
| 28                                               | 13.565                                    | 16.928 | 18.939 | 22.657 | 27.336 | 32.62 | 37.92 | 41.34 | 48.28 |
| 30                                               | 14.953                                    | 18.493 | 20.599 | 24.478 | 29.336 | 34.80 | 40.26 | 43.77 | 50.89 |
| 40                                               | 22.164                                    | 26.509 | 29.051 | 33.660 | 39.335 | 45.62 | 51.80 | 55.76 | 63.69 |
| 50                                               | 27.707                                    | 34.764 | 37.689 | 42.942 | 49.335 | 56.33 | 63.17 | 67.50 | 76.15 |
| 60                                               | 37.485                                    | 43.188 | 46.459 | 52.294 | 59.335 | 66.98 | 74.40 | 79.08 | 88.38 |

## Question 31

M2L12 M7L6

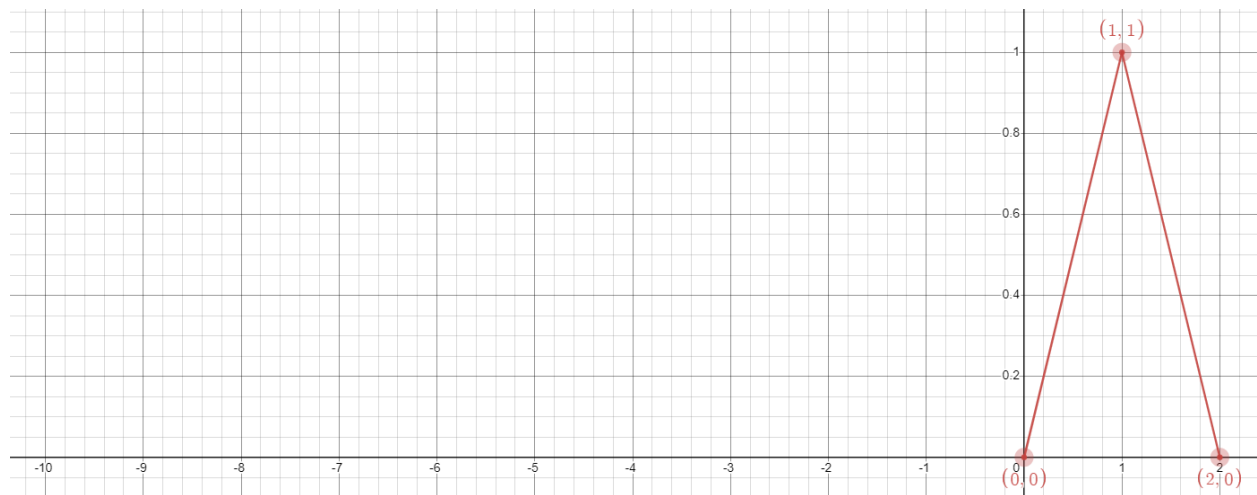
Review [Triangular Distribution Shifting and Scaling](#)

Recall that the parameterization of the triangular distribution is  $\text{Tria}(a, b, c)$  where  $a$  is the minimum,  $b$  is the most likely/mode, and  $c$  is the maximum.

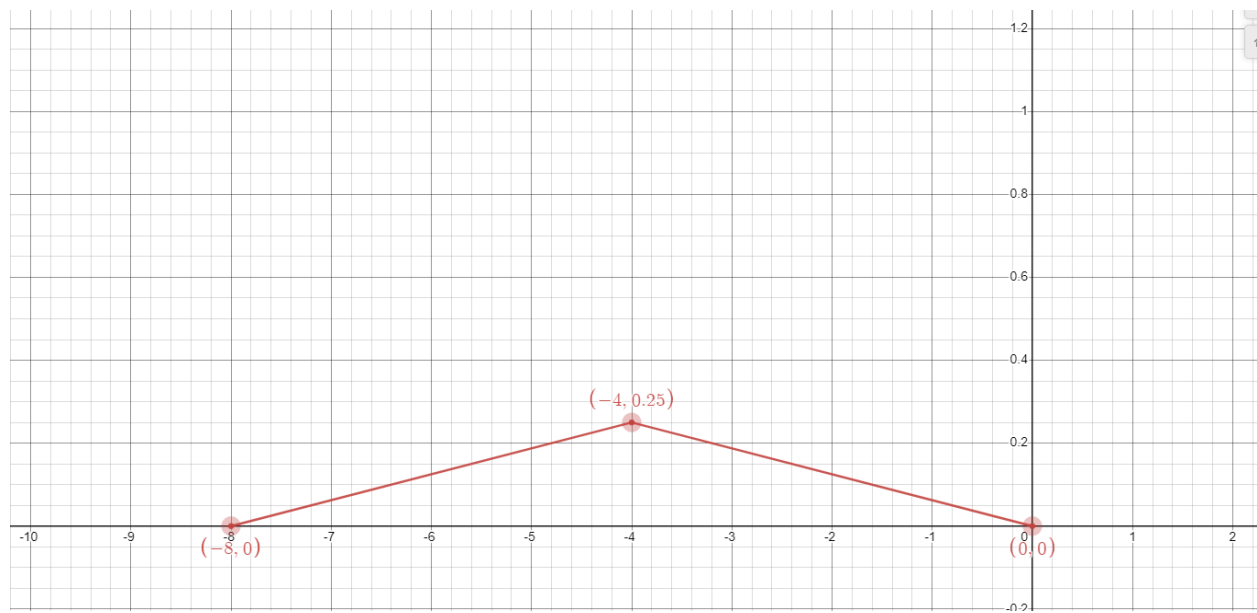
Note that these parameters ARE NOT the expected value and variance.

We can use a geometric argument to figure out what happens as we transform this random variable.

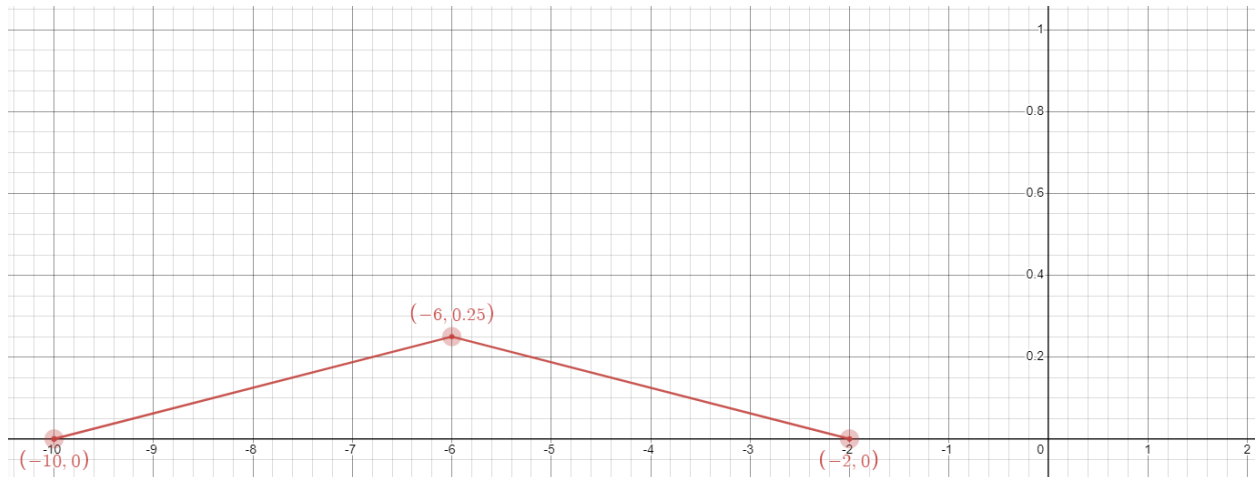
We know that  $U_1 + U_2 \sim \text{Tria}(0, 1, 2)$



When we multiply this random variable by  $-4$ , it will reflect the triangle across the y axis (due to the negative) and scale it by 4.



Subtracting 2 will translate it two units to the left.



The result is a  $\text{Tria}(-10, -6, -2)$  random variable.

## Question 32

### M7L7

(a) Ask a UGA student - This answer is not valid as it does not provide any computational or statistical technique for generating a random variate. Rather, it proposes asking a student, which doesn't ensure any theoretical or practical method.

(b) Convolution - Convolution involves adding up appropriate random variates. While this method is commonly used in generating some distributions, it is not typically used to generate Beta-distributed random variates. Thus, this choice is also invalid.

(c) Inversion - Inversion method, also known as the inverse transform sampling, works on the idea of transforming a uniform random variate in a way that the result has the required distribution. However, the Beta distribution does not have an easily computable inverse cumulative distribution function, making this approach more complex and less likely to be the best choice.

(d) Acceptance-Rejection - This method is one of the most commonly used for generating random variates when the inverse transform method is not practical. It involves generating a candidate from a distribution that is easy to sample from, and accepting it with a probability related to the ratio of the densities of the target and proposal distribution. Given the complexities associated with the Beta distribution, the Acceptance-Rejection method can be a valid and efficient choice.

(e) Composition - The composition method involves constructing the desired distribution from simpler distributions. While it is possible to generate a Beta distribution from simpler distributions, it's not the most straightforward or efficient method.

So, the best answer would be (d) Acceptance-Rejection, given that it is a common method for generating random variates from more complex distributions like the Beta distribution, when the inversion method is not feasible or straightforward.

## Question 33

On some of the practice exams, there were questions related to pop culture, which were considered bonus questions. Please do not stress over these, as they are meant to be fun and easy free points. If questions such as these appear on actual exams, you will get credit for them regardless.